Fatima Zohra BENTOUHAMI

Engineering Cycle in Software Engineering | Seeking a final-year internship (PFE): Full Stack Web/Mobile & AI

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Location: Morocco | Category B driving licence

PROFILE

Third-year software engineering student with hands-on **web, mobile and AI** experience: delivered back-end/full-stack modules and AI features during a 2-month remote internship. Rigorous, autonomous, interested in software development.

EDUCATION

- **Engineering Cycle in Computer Engineering** — National School of Applied Sciences, Al Hoceima (ENSAH), 2024 – 2027
- **Preparatory Classes for Engineering Schools (CPGE MP, Mathematics-Physics)**, 2022 – 2024
- **Baccalaureate in Physical Sciences, French option** — Distinction (Très Bien), 2022

PROFESSIONAL EXPERIENCE

PFA Intern, full-stack web & mobile development — QS Administration (Netherlands, remote) *07/2026 – 09/2026*

PFA internship (end-of-year project): web and mobile accounting management platform for clients and accountants

- **Delivered 9 back-end/full-stack modules in 2 months**: invoices, quotes, payments, purchases, banking, accounting entries, VAT, tasks, messaging.
- **Analyzed the existing application** (accounting workflows, Prisma schema, authentication, authorizations), then managed the PostgreSQL database via Prisma.
- **Integrated 8 server-side AI features**: invoice recognition, accounting categories, reply drafts, summaries, tax explanations, voice invoicing; “AI proposes, the accountant validates” principle.
- **Developed the mobile application** in React Native/Expo (**2 roles**: client, accountant): role-based navigation, document scanning, redesign of the interface and user experience (UI/UX), server-side access control.
- **Structured 2 applications (web + mobile, front-end and back-end) around 1 single REST API** and 1 monorepo (Next.js, React, React Native, TypeScript, PostgreSQL, Prisma).

PROJECTS

- **Ongoing development of a mobile app for CV analysis and interview preparation** (2026): job-offer matching, recommendations, AI-evaluated simulations. *Java Spring*, *PostgreSQL*, *RAG*, *AI*.
- **Designed an intelligent medical assistant in Darija** (2026): semantic search over **100,000+ medical Q&A pairs**, text and voice answer generation. *NLP*, *RAG*, *Deep Learning*.
- **Designed a course registration management platform** (2026): decomposition into independent services for scalability and maintainability. *Microservices*.
- **Developed a CNN model for early plant disease detection** (2026) from leaf images. *Deep Learning*, *CNN*.
- **Developed a Java road traffic simulation** (2025): real map, vehicle flow modeling and animated visualization. *Java*, *OpenStreetMap*.
- **Developed a teacher assignment management web application** (2024): automated course assignment, timetables and an administration interface. *PHP*, *MySQL*, *JavaScript*.

SKILLS

Programming languages: TypeScript, JavaScript, Python, Java, C, C++, SQL (Structured Query Language), object-oriented programming (OOP)

Front-end development: Next.js, React, React Native (Expo), HTML5/CSS

Back-end development: Spring Framework, Prisma, REST API (Representational State Transfer), API integration, microservices architecture, PHP

Databases: PostgreSQL, MySQL, SQLite, Prisma (ORM: object-relational mapping)

Artificial Intelligence (AI): Machine Learning, Deep Learning (convolutional neural networks CNN, natural language processing NLP), RAG (retrieval-augmented generation), LLM integration (large language models), Prompt Engineering

DevOps and tools: Git/GitHub, Docker, Linux, cloud computing

ERP (Enterprise Resource Planning): Odoo

Project management: Scrum, Agile methods

LANGUAGES & SOFT SKILLS

Languages: Arabic (mother tongue), French (professional), English (technical)

Soft skills: Rigor, autonomy, analytical thinking, teamwork, problem solving